Gaussian Elimination by using MATLAB

Step 1

```
% Augmented Matrix [A | b]


A = [2 1 3 5; 4 5 6 16; 6 7 8 21];
 disp('Initial Augmented Matrix:’)
disp(A)
```

Step 2

```
disp('=====================================‘)
disp('Step 1: First Pivot’)

% Pivot
p1 = A(1,1);
fprintf('Pivot p1 = a11 = %.2f\n', p1);
```

Step 3

```
% Multipliers
m21 = A(2,1)/p1;
m31 = A(3,1)/p1;
fprintf('Multiplier m21 = %.2f\n', m21);
fprintf('Multiplier m31 = %.2f\n', m31);
```

Step 4

```
% Row Operations
A(2,:) = A(2,:) - m21*A(1,:);
A(3,:) = A(3,:) - m31*A(1,:);

disp('Matrix after Step 1:') disp(A)
```

Step 5

```
disp('=====================================‘)
disp('Step 2: Second Pivot’)

% Pivot
p2 = A(2,2);
fprintf('Pivot p2 = a22 = %.2f\n', p2);
```

Step 6

```
% Multiplier
m32 = A(3,2)/p2;
fprintf('Multiplier m32 = %.2f\n', m32);

% Row Operation
A(3,:) = A(3,:) - m32*A(2,:);

disp('Matrix after Step 2:') disp(A)
```

Step 7

```
disp('==========================================‘)
disp('Step 3: Third Pivot’)

 % Pivot
p3 = A(3,3);

fprintf('Pivot p3 = a33 = %.2f\n', p3);
 disp('No row operations are needed.’)
disp('Upper Triangular Matrix:') disp(A)
```

Step 8

```
%% ---------------- Back Substitution ----------------
disp('=====================================‘)
disp('Back Substitution’)

z = A(3,4)/A(3,3);
y = (A(2,4)-A(2,3)*z)/A(2,2);
x = (A(1,4)-A(1,2)*y-A(1,3)*z)/A(1,1);

fprintf('x = %.2f\n',x);
fprintf('y = %.2f\n',y);
fprintf('z = %.2f\n',z);
```